



Software Process Improvement –SEPI-471



PROCESS MODELS FOR SPI- CMMI

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Agenda

- Process Models for SPI
 - Capability Maturity Model Integration (CMMI)

Capability Maturity Model Integration (CMMI)



- **CMMI Representation**
 - Staged
 - Continuous

staged representation

The staged representation is concerned with selecting multiple process areas to improve within a maturity level; whether individual processes are performed or incomplete is not the primary focus. Therefore, the name “Initial” is given to the staged representation starting point.

CMMI-Staged

- Uses a pre-defined set of process areas to define improvement path.
- provides a sequence of improvements, where each part in the sequence serves as a foundation for the next.
- an improved path is defined by maturity level.
- maturity level describes the maturity of processes in organization.
- Staged CMMI representation allows comparison between different organizations for multiple maturity levels.

CMMI-Staged

- In CMMI with staged representation, there are five maturity levels

CMMI-Staged

Maturity level 1 : Initial

1. processes are poorly managed or controlled.
2. unpredictable outcomes of processes involved.
3. ad hoc and chaotic approach used.
4. No KPAs (Key Process Areas) defined.
5. Lowest quality and highest risk.

CMMI-Staged

Maturity level 2 : Managed

1. requirements are managed.
2. processes are planned and controlled.
3. projects are managed and implemented according to their documented plans.
4. This risk involved is lower than Initial level, but still exists.
5. Quality is better than Initial level.

CMMI-Staged

Maturity level 3 : Defined

1. processes are well characterized and described using standards, proper procedures, and methods, tools, etc.
2. Medium quality and medium risk involved.
3. Focus is process standardization.

CMMI-Staged

Maturity level 4 : Quantitatively managed

1. quantitative objectives for process performance and quality are set.
2. quantitative objectives are based on customer requirements, organization needs, etc.
3. process performance measures are analyzed quantitatively.
4. higher quality of processes is achieved.
5. lower risk

CMMI-Staged

Maturity level 5 : Optimizing

1. continuous improvement in processes and their performance.
2. improvement has to be both incremental and innovative.
3. highest quality of processes.
4. lowest risk in processes and their performance.

CMMI-Continuous

Capability level 0 : Incomplete

1. incomplete process – partially or not performed.
2. one or more specific goals of process area are not met.
3. No generic goals are specified for this level.
4. this capability level is same as maturity level 1.

CMMI-Continuous

Capability level 1 : Performed

process area's specific goals are met (it's "performed"), but it might still be inconsistent or reliant on individuals, unlike the "Initial" maturity level where things are chaotic; Level 1 indicates basic tasks happen, but quality, cost, and schedule aren't predictable

CMMI-Continuous

Capability level 2 : Managed

1. process is planned, monitored and controlled.
2. managing the process by ensuring that objectives are achieved.
3. objectives are both model and other including cost, quality, schedule.
4. actively managing processing with the help of metrics.



CMMI-Continuous

Capability level 3 : Defined

a defined process is managed and meets the organization's set of guidelines and standards.

focus is process standardization.

CMMI-Continuous

Capability level 4 : Quantitatively Managed

1. process is controlled using statistical and quantitative techniques.
2. process performance and quality is understood in statistical terms and metrics.
3. quantitative objectives for process quality and performance are established.

CMMI-Continuous

Capability level 5 : Optimizing

1. focuses on continually improving process performance.
2. performance is improved in both ways – incremental and innovation.
3. emphasizes on studying the performance results across the organization to ensure that common causes or issues are identified and fixed.

CMMI Services in Pakistan

Right Track Technologies Pvt. Ltd.:

- **CMMI Partner:** Right Track Technologies Pvt. Ltd. (RTTPL), based in Islamabad, is a CMMI Institute Partner. They provide consulting and support services in areas such as project management, process improvement, process definition, measurement, and risk management.

<https://cmmiinstitute.com/partners/directory/right-track-technologies-pakistan>

Catalytic Consulting:

- **CMMI Appraisal Services:** Catalytic Consulting offers CMMI appraisal and consulting services in Pakistan.
- <https://www.catalytic.com/>

CMMI Services in Pakistan

GAK Technologies:

- **CMMI Consultancy:** GAK Technologies is a professional certification and consulting firm offering CMMI certification services across major cities in Pakistan, including Karachi, Lahore, and Islamabad. They provide a 100% success guarantee for CMMI certification. gaktechnologies.com

<https://gaktechnologies.com>

<https://gaktechnologies.com/cmmi-consultancy-in-pakistan-karachi.php>

CMMI Services in Pakistan

TopCertifier:

- **CMMI Consulting:** TopCertifier offers comprehensive support and expertise to assist companies in achieving CMMI certification in Pakistan. Their services include CMMI gap assessment, process improvement consulting, appraisal support, and training across cities like Lahore, Karachi, Islamabad, Faisalabad, Rawalpindi, and Peshawar.

<https://www.certification-consulting-pakistan.com/cmml-certification.html>

CMMI certified companies in Pakistan

1. Interactive Convergence:

- **CMMI Maturity Level:** Level 5
- **Details:** Interactive Convergence has been appraised at CMMI Level 5, the highest recognition in software quality management, positioning them among a select group of global organizations at this maturity level.

2. Zong 4G:

- **CMMI Maturity Level:** Level 3
- **Details:** Zong 4G, a leading telecom operator in Pakistan, celebrated its attainment of CMMI Maturity Level 3 in January 2024, underscoring their commitment to enhancing software development practices and delivering high-quality products.

CMMI certified companies in Pakistan

3. Avanza Solutions:

- **CMMI Maturity Level:** Level 2
- **Details:** Avanza Solutions' Product Development Unit has been appraised at CMMI Level 2, indicating their dedication to process improvement and effective performance.
avanzasolutions.com

4. Zong's Software Development & Automation Department:

- **CMMI Maturity Level:** Level 2
- **Details:** In December 2014, Zong's Software Development & Automation Department was appraised at CMMI Level 2, highlighting their dedication to process improvement and effective performance. breorder.com

Stages Vs Continuous CMMI

Staged CMMI gives **one maturity level** to the whole company (like Level 3 overall).
Continuous CMMI gives a **separate level** to each process (like Testing Level 2, Requirements Level 3).
Staged improves everything together step-by-step.
Continuous improves **only the areas you choose**.
Example: In staged, the company must reach all requirements for Level 4; in continuous, only the selected process needs to reach Level 4